
ENR 1.8 REGIONAL SUPPLEMENTARY PROCEDURES (DOC 7030)

1. SPECIAL PROCEDURES FOR IN-FLIGHT CONTINGENCIES

1.1 General Procedures

1.1.1 The following general procedures apply to both subsonic and supersonic aircraft and are intended as guidance only. Although all possible contingencies cannot be covered, they provide for cases of inability to maintain assigned level due to:

- a. weather;
- b. aircraft performance;
- c. pressurization failure; and
- d. problems associated with high-level supersonic flight.

1.1.2 The procedures are applicable primarily when rapid descent and/or turn-back or diversion to an alternate airport is required. The pilot's judgment shall determine the sequence of actions to be taken, taking into account specific circumstances.

1.1.3 If an aircraft is unable to continue flight in accordance with its ATC clearance, a revised clearance shall, whenever possible, be obtained prior to initiating any action, using a distress or urgency signal as appropriate.

1.1.4 If prior clearance cannot be obtained, an ATC clearance shall be obtained at the earliest possible time and, until a revised clearance is received, the pilot shall:

- a. if possible, deviate away from an organized track or route system;
- b. establish communications with and alert nearby aircraft by broadcasting, at suitable intervals: aircraft identification, flight level, aircraft position, (including the ATS route designator or the track code) and intentions on the frequency in use, as well as on frequency 121.5MHZ (or, as a back-up, the VHF inter-pilot air-to-air frequency 123.45MHZ);
- c. watch for conflicting traffic both visually and by reference to ACAS (if equipped); and
- d. turn on all aircraft exterior lights (commensurate with appropriate operating limitations).

1.2 Special procedures for subsonic aircraft requiring rapid descent, and/or turn-back or diversion to an alternate airport due to aircraft system malfunction or other contingencies

1.2.1 Initial action

1.2.1.1 If unable to comply with the provisions of paragraph 1.1.3 to obtain a revised ATC clearance, the aircraft should leave its assigned route or track by turning 90 degrees to the right or left whenever this is possible. The direction of the turn, where possible, be determined by the position of the aircraft relative to any organized route or track system e.g. whether the aircraft is outside, at the edge of, or within the system. Other factors to consider are the direction to the alternate airport, terrain clearance and the levels allocated to adjacent routes or tracks.

1.2.2 Subsequent action (RVSM airspace)

1.2.2.1 In RVSM airspace, an aircraft able to maintain its assigned flight level should turn to acquire and maintain in either direction a track laterally separated by 46KM (25NM) from its assigned route or track in multi-track system spaced at 93KM (50NM) or otherwise, at a distance which is the mid-point from the adjacent parallel route or track, and;

- a. if above FL410, climb or descend 300M (1000FT); or
- b. if below FL410, climb or descend 150M (500FT); or
- c. if at FL410, climb 300M (1000FT) or descend 150M (500FT).

1.2.2.2 An aircraft that is unable to maintain its assigned flight level should:

- a. initially minimize its rate of descent to the extent that it is operationally feasible;
- b. turn while descending to acquire and maintain in either direction a track laterally separated by 46KM (25NM) from its assigned route or track in a multi-track system spaced at 93KM (50NM) or otherwise, at a distance which is the mid-point from the adjacent parallel route or track; and
- c. for the subsequent flight level, select a level which differs from those normally used by 300M (1000FT) if above FL410, or by 150M (500FT) if below FL410.

1.2.3 Subsequent action (non-RVSM airspace)

1.2.3.1 In non-RVSM airspace, an aircraft able to maintain its assigned flight level should turn to acquire and maintain either direction a track laterally separated by 46KM (25NM) from its assigned route or track in a multi-track system spaced at 93KM (50NM) or otherwise, at a distance which is the mid-point from the adjacent parallel route or track, and:

- a. if above FL290, climb or descend 300M (1000FT); or
- b. if below FL290, climb or descend 150M (500FT); or
- c. if at FL290, climb 300M (1000FT) or descend 150M (500FT).

1.2.3.2 An aircraft that is unable to maintain its assigned level should:

- a. initially minimize its rate of descent to the extent that it is operationally feasible;
- b. turn while descending and maintain in either direction a track laterally separated by 46KM (25NM) from its assigned route or track in a multi-track system spaced at 93KM (50NM) or otherwise, at a distance which is the mid-point from the adjacent parallel route or track; and
- c. for the subsequent flight level, select a level which differs from those normally used by 300M (1000FT) if above FL290, or by 150M (500FT) if below FL290.

1.2.3.3 DIVERSION ACROSS THE FLOW OF ADJACENT TRAFFIC. Before diverting across the flow of adjacent traffic, the aircraft should climb above FL410 or descend below FL280 using the procedures specified in 1.2.1 or 1.2.2 or 1.2.3. However if the pilot is unable or unwilling to carry out a major climb or descent, the aircraft should be flown at a level defined in 1.2.2.1 or 1.2.3.1 until a revised ATC clearance is obtained.

1.2.3.4 EXTENDED RANGE OPERATIONS BY AEROPLANES WITH TWO-TURBINE POWER-UNITS (ETOPS) AIRCRAFT. If these contingency procedures are employed by a twin-engine aircraft as a result of an engine shutdown or a failure of an ETOPS critical system, the pilot should advise ATC as soon as practicable of the situation, reminding ATC of the type of aircraft involved and request expeditious handling.

1.3 Weather Deviation Procedures for oceanic-controlled airspace

1.3.1 General procedures

1.3.1.1 The following procedures are intended to provide guidance. All possible circumstances cannot be covered. The pilot's judgment shall ultimately determine the sequence of actions taken and ATC

shall render all possible assistance.

1.3.1.2 If the aircraft is required to deviate from track to avoid weather and prior clearance cannot be obtained, an ATC clearance shall be obtained at the earliest possible time. Until an ATC clearance is received, the aircraft shall follow the procedures detailed in 1.3.4 below.

1.3.1.3 The pilot shall advise ATC when weather deviation is no longer required, or when a weather deviation has been completed and the aircraft has returned to the centerline of its cleared route.

1.3.2 Obtaining priority from ATC when weather deviation is required

1.3.2.1 When the pilot initiates communications with ATC, rapid response may be obtained by stating "WEATHER DEVIATION REQUIRED" to indicate that priority is desired on the frequency and for ATC response.

1.3.2.2 The pilot still retains the option of initiating the communications using the urgency call "PAN PAN" (preferably spoken three times) to alert all listening parties to a special handling condition, which will receive ATC priority for issuance of a clearance or assistance.

1.3.3 Actions to be taken when controller-pilot communications are established

1.3.3.1 Pilot notifies ATC and requests clearance to deviate from track, advising, when possible, the extent of the deviation expected.

1.3.3.2 ATC takes one of the following actions:

- a. if there is no conflicting traffic in the horizontal dimension, ATC will issue clearance to deviate from track; or
- b. if there is conflicting traffic in the horizontal dimension, ATC separates aircraft by establishing vertical separation; or
- c. if there is conflicting traffic in the horizontal dimension and ATC is unable to establish appropriate separation, ATC shall:
 - i. advise the pilot unable to issue clearance for requested deviation; and
 - ii. advise the pilot of conflicting traffic; and
 - iii. request pilot's intentions

SAMPLE PHRASEOLOGY:

"UNABLE (requested deviation), TRAFFIC IS (call sign, position, altitude, direction), ADVISE INTENTIONS."

1.3.3.3 The pilot will take the following actions:

- a. advise ATC of intentions by the most expeditious means available; and
- b. comply with ATC clearance issued; or
- c. execute the procedures detailed in 1.3.4 (ATC will issue essential traffic information to all affected aircraft); and
- d. if necessary, establish voice communications with ATC to expedite dialogue on the situation.

1.3.4 Actions to be taken if a revised ATC clearance cannot be obtained

1.3.4.1 The provisions of this section apply to situations where the pilot has the need to exercise the authority of a pilot-in-command under the provisions of Annex 2, 2.3.1.

1.3.4.2 If a revised ATC clearance cannot be obtained and deviation from track is required to avoid weather, the pilot shall take the following actions:

- a. if possible, deviate away from an organized track or route system;
- b. establish communications with and alert nearby aircraft by broadcasting, at suitable intervals: aircraft identification, flight level, aircraft position (including the ATS route designator or the track code), and intentions (including the magnitude of the deviation expected) on the frequency in use, as well as on frequency 121.5MHZ (or, as a back-up, the VHF inter-pilot air-to-air frequency 123.45MHZ);
- c. watch for conflicting traffic both visually and by reference to ACAS (if equipped);

NOTE: If, as a result of actions taken by 1.3.4.2 b) and c), the pilot determines that there is another aircraft at or near the same flight level with which a conflict may occur, then the pilot is expected to adjust the path of the aircraft, as necessary, to avoid conflict.

- d. turn on all aircraft exterior lights (commensurate with appropriate operating limitations);
- e. for deviations of less than 19KM (10NM), aircraft should remain at the level assigned by ATC;
- f. for deviations of greater than 19KM (10NM), when the aircraft is approximately 19KM (10NM) from track initiate a level change based on the criteria in Table 1.
- g. when returning to track, be at its assigned level, when the aircraft is within approximately 19KM (10NM) of center line; and
- h. if contact was not established prior to deviating continue to attempt to contact ATC to obtain a clearance. If contact was established, continue to keep ATC advised of intentions and obtain essential traffic information.

Table 1

Route center line Track	Deviations > 19KM (10NM)	Level change
EAST 000 - 179 magnetic	LEFT RIGHT	Descend 90M (300FT) Climb 90M (300FT)
WEST 180 - 359 magnetic	LEFT RIGHT	Climb 90M (300FT) Descend 90M (300FT)

2. REDUCED VERTICAL SEPARATION MINIMUM (RVSM) POLICY AND PROCEDURES IN THE MANILA FIR

IMPLEMENTATION OF REVISED FLIGHT LEVEL ARRANGEMENTS IN THE WESTERN PACIFIC/SOUTH CHINA SEA AREA

2.1 Introduction

2.1.1 During 2002, in two stages during February and October respectively, the States of the ICAO Asia Pacific Region within the Western Pacific/South China Sea (WPAC/SCS) area implemented RVSM operations using a modified single alternate Flight Level Orientation Scheme (FLOS), with a complementary Flight Level Allocation Scheme (FLAS) for ATC flight level assignment.

2.1.2 To assist expeditious traffic handling, ATC operational arrangements termed 'no pre-departure coordination' (i.e. 'No-PDC') procedures are mutually agreed between affected ATC Area Control Centers (ACCs). Using No-PDC procedures mean the initial flight level for departing flights is allocated in accordance with the pre-agreed FLAS without real time flight level coordination being undertaken between adjacent ACCs. After departure, other flight levels may be available subject to prior coordination between ACCs to agree alternative flight levels for assignment.

2.1.3 Subsequent to the commencement of RVSM operations in the WPAC/SCS area, implementation of RVSM has continued in airspaces surrounding the WPAC/SCS area but utilizing a single alternate FLOS in accordance with the Tables of Cruising Levels contained in the Table "RVSM-FEET" of Appendix 3 of ICAO Annex 2 - Rules of the Air. This resulted in a need to continuously transition the flight levels of many flights entering and leaving the WPAC/SCS area between the modified single alternate FLOS in the WPAC/SCS area and the single alternate FLOS in surrounding RVSM areas.

2.2 Implement revised FLOS and FLAS in WPAC/SCS area

2.2.1 In order to minimize flight level transition requirements for flights entering and leaving the WPAC/SCS area, affected States working under the auspices of the ICAO WPAC/SCS RVSM Scrutiny Working Group (WPAC/SCS RSG) will implement revised flight level arrangements for the WPAC/SCS area in association with the 3 July 2008 AIRAC date.

2.2.2 With effect from 0807022100, simultaneous and permanent implementation of the following flight level arrangements in the WPAC/SCS area will occur:

- a. a single alternate FLOS (i.e. 'east odd flight levels, west even flight levels') in compliance with the Table "RVSM-FEET" of Appendix 3 of Annex 2 and in accordance with the FLOS in surrounding areas;
- b. special high capacity arrangements for six unidirectional parallel routes (L642, M771, N892, L625, N884 and M767) that involve managed use of odd and even flight levels in the same direction of flight; and
- c. an associated FLAS agreed between affected ACCs to facilitate ATC 'No-PDC' operations.

2.2.3 Details of the flight level arrangements to be implemented, including those applicable to Large Scale Weather Deviation (LSWD), have been included in Table 2.2.5.

2.2.4 FLAS with adjacent FIRs.

FIR	No Pre-Departure Coordination (PDC) Flight Levels	ATS / RNAV Route	Flight Level Transition
Manila - Fukuoka	FL310, 320, 350, 360, 390, 400	L625 MEVIN B462	Fukuoka ACC will transition FL320, 360, 400 to ODD levels after MEVIN.
		LAO B462	
	FL290, 310, 330, 350, 370, 390, 410	N884	Nil.
Fukuoka - Manila	FL300, 340, 380	B462 LAO	Nil.
	FL300, 320, 340, 360, 380, 400	A582	
Manila - Fukuoka	FL290, 310, 330, 350, 370, 390, 410	A590	Nil.
		G578	
Fukuoka - Manila	FL300, 320, 340, 360, 380, 400	A590	Nil.
		G578	
Manila - Oakland	FL290, 310, 330, 350, 370, 390, 410	M501, G467, R337	Nil.
	FL310, 350, 390	A339	
Oakland - Manila	FL300, 320, 340, 360, 380, 400	M501, G467, R337	Nil.
	FL320, 360, 400	A339	
Manila - Ujung Pandang	FL290, 310, 330, 350, 370, 390, 410	A461, R590, B472, B473, B462	Nil.
	FL320, 360, 400	G578	
	FL320, 360, 400	A339	
Ujung Pandang - Manila	FL300, 340, 380	A461, R590, B472, B473, B462	Nil.
	FL350, 390	G578	
	FL310, 350, 390	A339	
Manila - Kota Kinabalu	FL290, 330, 370, 410	M754	Nil.
	FL320, 360, 400	M646	
	FL320, 360, 400	A341	
Kota Kinabalu - Manila	FL300, 340, 380	M754	Nil.
	FL310, 350, 390	M646	
	FL310, 370	A341	
Manila - Singapore	FL310, 320, 350, 360, 390, 400	M767	Manila ACC will transition FL300, 340, 380 to FL310, 350, 390 before JOM.
Singapore - Manila	FL310, 320, 350, 360, 390, 400	N884	Manila ACC will transition FL320, 360, 400 to ODD levels between LULBU and LUBAN.
	FL300, 380	M772	Nil.
Manila - Ho Chi Minh	FL310, 320, 350, 360, 390, 400	N892	Nil.
	FL280, 340	L628, M765	
Ho Chi Minh - Manila	FL310, 320, 350, 360, 390, 400	L625	Nil.
	FL330, 370, 410	L628	
	FL290, 370	M765	

FIR	No Pre-Departure Coordination (PDC) Flight Levels	ATS / RNAV Route	Flight Level Transition
Manila - Hong Kong	FL300, 340, 380	A461, A583, M501/A461	Nil.
	FL300, 380	M772	
Hong Kong - Manila	FL290, 330, 370, 410	A461, A583, A461/M501	Nil.
Manila - Taipei	FL290, 310, 330, 350, 370, 390, 410	L625/M646	Manila ACC will transition FL320, 360, 400 to ODD levels before POTIB.
	FL290, 310, 330, 350, 370, 390, 410	LAO M646	Nil.
Taipei - Manila	FL310, 320, 350, 360, 390, 400	N892	Taipei ACC will transition FL300, 340, 380 to FL310, 350, 390 before KABAM.
	FL300, 340, 380	M646	Nil.

2.2.5 FLAS for LSWD in Western Pacific/South China Sea area as applicable by:

Fukuoka, Japan ATMC, Ho Chi Minh, Hong Kong, Manila, Sanya, Singapore and Taipei ACCs.

Flight Level Allocation (LSWD)	ATS Route and Direction of Flight								
	N892	L625	N884 LAXOR- LUBAN	N884 CAB-LEBIX	M767	A582/B462		A590	
	SW	NE	NE	NE	SW	E	W	E	W
410				✓					
400	✓				✓				✓
390		✓	✓			✓		✓	
380							✓		
370				✓					
360	✓				✓				✓
350		✓	✓			✓		✓	
340							✓		
330				✓					
320	✓				✓				✓
310		✓	✓			✓		✓	
300							✓		
290				✓					

2.3 Airworthiness and Operational Approval and Monitoring

2.3.1 **Approval Date.** Operator/aircraft approval by 1 January 2002 will enable air traffic service providers to plan for orderly RVSM implementation.

2.3.2 **Approval Process.** Operators must obtain airworthiness and operational approval from the State of Registry or State of the Operator, as appropriate, to conduct RVSM operations.

2.3.3 **Aircraft Monitoring.** Operators are required to participate in the RVSM aircraft-monitoring program. The Monitoring Agency for Asia Region (MAAR) will process the results of monitoring.

- 2.3.3.1 Monitoring accomplished for other regions can be used to fulfill the monitoring requirements for the Asia/Pacific region. The MAAR will coordinate with other monitoring agencies to access this information. For monitoring services in the Asia/Pacific region, operators should contact the MAAR monitoring contractor as follows:

Fax: +66-2-297-8155
Email: maar@aerodhai.co.th

2.4 Airborne Collision Avoidance System II (ACAS II)

- 2.4.1 Any airborne collision avoidance system installed on an aircraft shall be approved by the Authority.
- 2.4.2 Each person operating an aircraft equipped with an airborne collision avoidance system shall have that system on and operating.
- 2.4.3 No person may operate a turbine engine airplane for which the individual airworthiness certificate was first issued after 24 November 2005 with a maximum certificated take-off mass in excess of 15000 KG or authorized to carry more than 30 passengers, unless it is equipped with an ACAS II.
- 2.4.4 No person may operate a turbine engine airplane for which the individual airworthiness certificate was first issued after 1 January 2008 with a maximum certificated take-off mass in excess of 5700 KG but not exceeding 15000 KG or authorized to carry more than 19 passengers, unless it is equipped with an ACAS II.
- 2.4.5 An airborne collision avoidance system shall operate as designed.
- 2.4.6 No person may operate a turbine powered airplane with a maximum certificated take-off mass in excess of 5700 KG or authorized to carry more than 19 passengers, unless it is equipped with an ACAS II.
- 2.4.7 After 1 January 2017, all Airborne Collision Avoidance System (ACAS) units shall comply with requirements of 2.4 through the implementation of Traffic Alert and Collision Avoidance System (TCAS) Version 7.1.

2.5 In-flight Procedures within RVSM Airspace

- 2.5.1 Before entering RVSM airspace, the pilot should review the status of required equipment. The following equipment should be operating normally:
- two primary altimetry systems;
 - one automatic altitude-keeping device; and
 - one altitude-alerting device.
- 2.5.2 The pilot must notify ATC whenever the aircraft:
- is no longer RVSM compliant due to equipment failure; or
 - experiences loss of redundancy of altimetry systems; or
 - encounters turbulence that affects the capability to maintain flight level.
- 2.5.3 Transition between FL's. During cleared transition between levels, the aircraft should not overshoot or undershoot the assigned FL by more than 150 FT (45 M).
- 2.5.4 Pilot Level Call. Except in an ADS or radar environment, pilots shall report reaching any altitude assigned within RVSM airspace.
- 2.5.5 Contingency Procedures. ENR 1.8 items 1.1, 1.2 & 1.3 contain procedures for in-flight

contingencies that have been updated for RVSM operations. The contingency procedures in ENR 1.8 items 1.1 to 1.2 and the offset procedures in ENR 1.8 item 2.6 should be applied in oceanic operations. The weather deviation procedures in ENR 1.8 item 1.3 may be applied in all airspace in the region.

2.6 Special Procedures to Mitigate Wake Turbulence Encounters and Distracting Aircraft System Alerts in the Oceanic Airspace of the Manila FIR

2.6.1 The following special procedures are applicable to mitigate wake turbulence or distracting aircraft system alerts (e.g. ACAS, Ground Proximity Warning System (GPWS)) in Asia and Pacific airspace where RVSM is applied.

Note: In the contingency circumstances below, ATC will not issue clearances for lateral offsets and will not normally respond to actions taken by the pilots.

2.6.2 An aircraft that encounters wake vortex turbulence or experiences distracting aircraft system alerts shall notify ATC and request a flight level, track or speed change to avoid the condition. However, in situations where such change is not possible or practicable, the pilot may initiate the following temporary lateral offset procedure with the intention of returning to center line as soon as practicable:

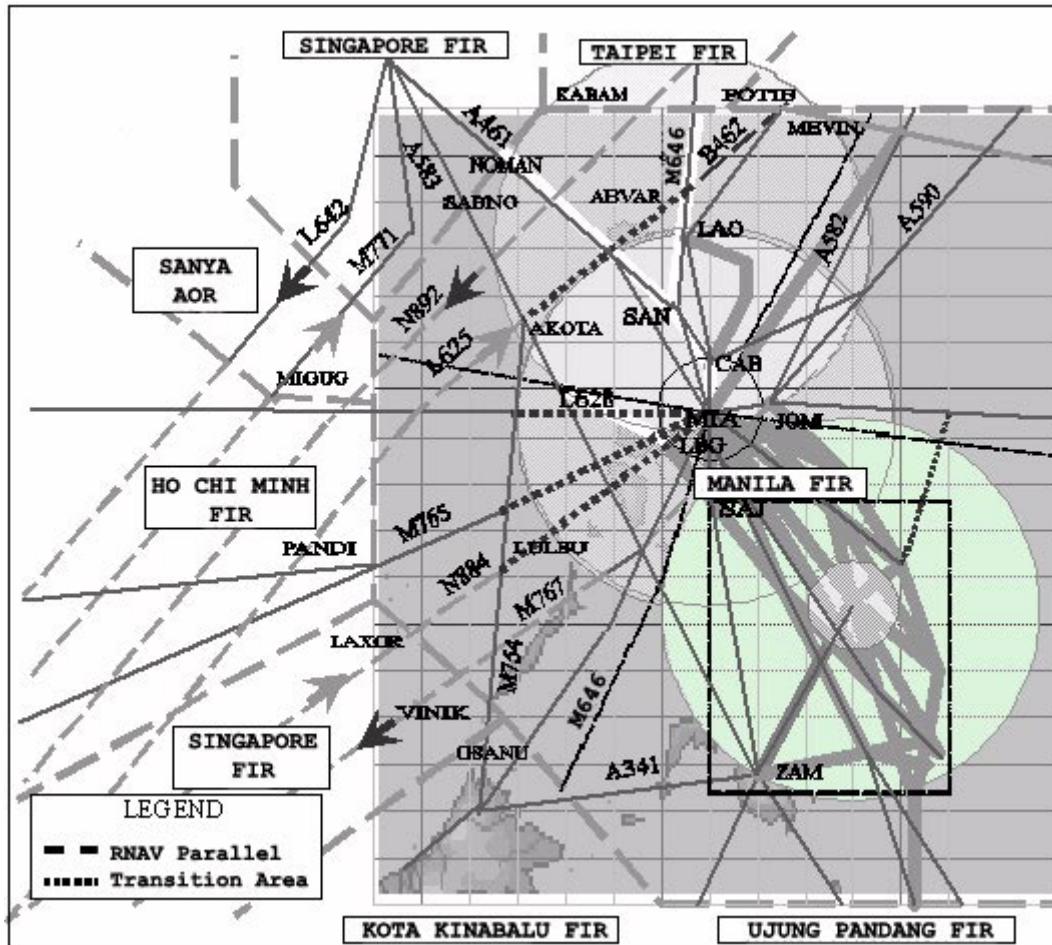
- a. the pilot should establish contact with other aircraft, if possible, on the appropriate VHF inter-pilot air to air frequency, 123.45 MHZ; and
- b. one (or both) aircraft may initiate lateral offset(s) not to exceed 2 NM from the assigned track, provided that:
 - i. as soon as practicable to do so, the offsetting aircraft will notify ATC that temporary lateral offset action has been taken and specify the reason for doing so (ATC will not normally respond); and
 - ii. the offsetting aircraft will notify ATC when re-established on assigned route(s) or track(s) (ATC will not normally respond).

2.7 Transition Areas

2.7.1 Below are the identified transition areas (Figure 1. RVSM Transition Area Chart) and procedures within the Manila FIR:

VERTICAL Transition Areas	Flights Concerned
After Manila VOR	Westbound overflights coming via A590, A582, G467, joining RNAV routes L628, M765.
Upon radar contact between LULBU and LUBAN	Eastbound overflights via RNAV route N884, joining A590, A582, G467 to ENDAX.
Upon radar contact between AKOTA and ABVAR	Eastbound overflights on RNAV route L625 joining M646, MEVIN B462.
Within Radar Coverage	Other flights not covered above.

Figure 1. RVSM Transition Area Chart



2.8 Flight Planning Requirements

2.8.1 Unless special arrangement is made as detailed below, RVSM approval is required for operators and aircraft to operate within designated RVSM airspace. The operator must determine that the appropriate State Authority has granted them RVSM operational approval and they will meet the RVSM requirements for the filed route of flight and any planned alternate routes. The letter "W" shall be inserted in item 10 (Equipment) of the ICAO standard flight plan to indicate that both the aircraft and operator are RVSM approved.

2.9 Procedures for Operation of Non-RVSM Compliant Aircraft in RVSM Airspace

2.9.1 **Flight priority.** RVSM approved aircraft will be given priority for level allocation over non-RVSM approved aircraft.

2.9.2 **Vertical separation applied.** The vertical separation minimum between non-RVSM aircraft operating in the RVSM stratum and all other aircraft is 2000 FT.

2.9.3 **Phraseology.** Non-RVSM compliant aircraft operating in RVSM airspace should use the phraseology contained in Table 2.9.3-1.

Table 2.9.3-1 - Phraseology Related to RVSM Operations	
Message	Phraseology
Controller-pilot phraseology:	
a. For a controller to ascertain the RVSM approval status of an aircraft.	(call sign) CONFIRM RVSM APPROVED
b. For a pilot to report non-RVSM approval status:	NEGATIVE RVSM
i. on the initial call on any frequency within the RVSM airspace (controllers shall provide a read back with this same phrase); ii. in all requests for flight level changes pertaining to flight levels within the RVSM airspace; and iii. in all read-backs to flight level clearances pertaining to flight levels within the RVSM airspace. Additionally, except for State aircraft, pilots shall include this phrase to read back flight level clearances involving the vertical transit through FL290 or FL410.	
c. For a pilot to report RVSM approval status.	AFFIRM RVSM
d. For a pilot of a non-RVSM approved State aircraft to report non-RVSM approval status, in response to the phrase (call sign) CONFIRM RVSM APPROVED.	NEGATIVE RVSM STATE AIRCRAFT
e. Denial of clearance into the RVSM airspace:	(call sign) UNABLE CLEARANCE INTO RVSM AIRSPACE, MAINTAIN [or DESCEND TO, or CLIMB TO] FLIGHT LEVEL (number)
f. For a pilot to report when severe turbulence affects the aircraft's capability to maintain the height-keeping requirements for RVSM.	UNABLE RVSM DUE TURBULENCE
g. For a pilot to report that the aircraft's equipment has degraded enroute below that required for flight within the RVSM airspace. (This phrase is to be used to convey both the initial indication of the non-MASPS compliance, and henceforth, on initial contact on all frequencies within the lateral limits of the RVSM airspace until such time as the problem ceases to exist, or the aircraft has exited the RVSM airspace).	UNABLE RVSM DUE EQUIPMENT
h. For a pilot to report the ability to resume operations within the RVSM airspace after an equipment or weather-related contingency.	READY TO RESUME RVSM
i. For a controller to confirm that an aircraft has regained its RVSM approval status, or to confirm that the pilot is ready to resume RVSM operations.	REPORT ABLE TO RESUME RVSM
Coordination between ATS units:	
a. To verbally supplement an automated estimate message exchange which does not automatically transfer Item 18 flight plan information.	NEGATIVE RVSM or NEGATIVE RVSM STATE AIRCRAFT [as applicable]
b. To verbally supplement estimate messages of non-RVSM approved aircraft.	NEGATIVE RVSM or NEGATIVE RVSM STATE AIRCRAFT [as applicable]
c. To communicate the cause of a contingency relating to an aircraft that is unable to conduct RVSM operations due to severe turbulence or other severe weather-related phenomenon [or equipment failure, as applicable].	UNABLE RVSM DUE TURBULENCE [or EQUIPMENT, as applicable]

Example 1: A non-RVSM approved aircraft, maintaining FL260, subsequently requests a climb to FL320.

Pilot: (call sign) REQUEST FL320, NEGATIVE RVSM

Controller: (call sign) CLIMB TO FL320

Pilot: (call sign) CLIMB TO FL320, NEGATIVE RVSM

Example 2: A non-RVSM approved aircraft, maintaining FL260, subsequently requests a climb to FL430.

Pilot: (call sign) REQUEST FL430, NEGATIVE RVSM

Controller: (call sign) CLIMB TO FL430

Pilot: (call sign) CLIMB TO FL430, NEGATIVE RVSM

Example 3: A non-RVSM approved aircraft, maintaining FL360, subsequently requests a climb to FL380.

Pilot: (call sign) REQUEST FL380, NEGATIVE RVSM

Controller: (call sign) CLIMB TO FL380

Pilot: (call sign) CLIMB TO FL380, NEGATIVE RVSM

Example 4: A non-RVSM approved civil aircraft maintaining FL280, subsequently requests a climb to FL320.

Pilot: (call sign) REQUEST FL320, NEGATIVE RVSM

Controller: (call sign) UNABLE CLEARANCE INTO RVSM AIRSPACE, MAINTAIN FL280

2.9.4 **Continuous climb/descent of non-compliant aircraft through RVSM airspace.** Non-RVSM compliant aircraft may be cleared to climb to and operate above FL290 or descend to and operate below FL410 provided that they:

- a. Do not climb or descend at less than the normal rate for the aircraft; and
- b. Do not level off at an intermediate level while passing through the RVSM stratum.

2.9.5 **Special coordination procedures for cruise operation of non-RVSM compliant aircraft in RVSM airspace.** Non-RVSM compliant aircraft may not flight plan between FL290 and FL410 inclusive within RVSM airspace. After special coordination as detailed in 2.9.5.1 below, the following non-RVSM aircraft may flight plan at RVSM flight levels in the RVSM stratum:

- a. The aircraft is being initially delivered to the State of Registry or Operator; or
- b. The aircraft was formally RVSM approved but has experienced an equipment failure and is being flown to a maintenance facility for repair in order to meet RVSM requirements and/or obtain approval; or
- c. The aircraft is transporting a spare engine mounted under the wing; or
- d. The aircraft is being utilized for mercy or humanitarian purposes; or
- e. State aircraft (those aircraft used in military, custom and police services shall be deemed state aircraft).

Note: The procedures are intended exclusively for the purposes indicated and not as a means to circumvent the normal RVSM approval process.

| 2.9.5.1 The assignment of cruising levels to non-RVSM compliant aircraft listed in items 2.9.5 (a) to (e) shall

be subjected to ATC clearance. Aircraft operators shall include the “STS/Category of operations (i.e., FERRY / HUMANITARIAN / MILITARY / CUSTOMS / POLICE) / NON-RVSM COMPLIANT” in Field 18 of the ICAO Flight Plan.

2.9.5.2 Where necessary, the Air Traffic Control Center may be contacted as follows:

Manila Area Control Center
Phone: (632) 8672-7770 (Supervisor)
(632) 8672-7784 (Facility Chief)
AFTN: RPHIATMS

2.10 Delivery flights for aircraft that are RVSM compliant on delivery

2.10.1 An aircraft that is RVSM compliant on delivery may operate in RVSM airspace provided that the crew is trained on RVSM policies and procedures applicable in the airspace and the responsible State issues the operator a letter of authorization approving the operation. State notification to the MAAR should be in the form of a letter, e-mail or fax documenting the one-time flight. The planned date of the flight, flight identification, registration number, and aircraft type/series should be included.

2.11 Procedures for suspension of RVSM

2.11.1 Air traffic services will consider suspending RVSM procedures within affected areas of the Manila FIR when there are pilot reports of greater than moderate turbulence. Within areas where RVSM procedures are suspended, the vertical separation minimum between all aircraft will be 2000 FT. In the assignment of levels, aircraft operating on the four major RNAV routes (via N892, L625, N884 and M767) would have priority. Aircraft operating on routes that cross the four major routes would be assigned levels, subject to coordination with the affected FIRs.

2.12 Guidance for pilots and controllers for actions in the event of aircraft system malfunction or turbulence greater than moderate

2.12.1 Contingency Scenarios

2.12.1.1 Table 2.12.1-1 and 2.12.1-2 summarizes pilot actions to mitigate the potential conflict with other aircraft in certain contingency scenarios. They should be reviewed in conjunction with the expanded contingency scenarios detailed in ENR 1.8 items 2.5 to 2.6 and ENR 1.8 item 1, which contain additional technical and operational detail.

Table 2.12.1-1 - Contingency * Scenario # 1	
The pilot is: 1) unsure of the vertical position of the aircraft due to the loss or degradation of all primary altimetry systems, or 2) unsure of the capability to maintain cleared flight level (CFL) due to turbulence or loss of all automatic altitude control systems.	
The Pilot should:	ATC can be expected to:
Maintain CFL while evaluating the situation.	
Watch for conflicting traffic both visually and by reference to ACAS, if equipped; If considered necessary, alert nearby aircraft by 1. making maximum use of exterior lights; 2. broadcasting position, FL, and intentions on 121.5 MHZ (as a back-up, the VHF inter-pilot air-to-air frequency, 123.45 MHZ, may be used).	

Table 2.12.1-1 - Contingency * Scenario # 1	
Notify ATC of the situation and intended course of action. Possible courses of action include: - maintaining the CFL and route provided that ATC can provide lateral, longitudinal or conventional vertical separation. - requesting ATC clearance to climb above or descend below RVSM airspace if the aircraft cannot maintain CFL and ATC cannot establish adequate separation from other aircraft. - executing the contingency maneuver shown in ENR 1.8 subsections 1.2 and 1.3 to offset from the assigned track and FL, if ATC clearance cannot be obtained and the aircraft cannot maintain CFL.	Obtain the pilot's intentions and pass essential traffic information. - If the pilot intends to continue in RVSM airspace, assess traffic situation to determine if the aircraft can be accommodated through the provision of lateral, longitudinal, or conventional vertical separation, and if so, apply the appropriate minimum. - If the pilot requests clearance to exit RVSM airspace, accommodate expeditiously, if possible. - If adequate separation cannot be established and it is not possible to comply with the pilot's request for clearance to exit RVSM airspace, advise the pilot of essential traffic information, notify other aircraft in the vicinity and continue to monitor the situation. - Notify adjoining ATC facilities/sectors of the situation.

Table 2.12.1-2 - Contingency * Scenario # 2
There is a failure or loss of accuracy of one primary altimeter system (e.g., greater than 200 FT difference between primary altimeters.)
The Pilot should:
Cross check standby altimeter, confirm the accuracy of a primary altimeter system and notify ATC of the loss of redundancy. If unable to confirm primary altimeter system accuracy, follow pilot actions listed in the preceding scenario.

2.12.2 Expanded equipment failure and turbulence encounter scenarios

2.12.2.1 Operators may consider using Table 2.12.2-1 to 2.12.2-5 in training programs.

Table 2.12.2-1 - Contingency * Scenario # 1	
All automatic altitude control systems fail (e.g., Automatic Altitude Hold).	
The Pilot should:	ATC can be expected to:
Initially	
Maintain CFL.	
Evaluate the aircraft's capability to maintain altitude through manual control.	
Subsequently	
Watch for conflicting traffic both visually and by reference to ACAS, if equipped. If considered necessary, alert nearby aircraft by - making maximum use of exterior lights; - broadcasting position, FL, and intentions on 121.5 MHZ (as a back-up, the VHF inter-pilot air-to-air frequency, 123.45 MHZ, may be used)	

Table 2.12.2-1 - Contingency * Scenario # 1	
Notify ATC of the failure and intended course of action. Possible courses of action include: - maintaining the CFL and route, provided that the aircraft can maintain level. - requesting ATC clearance to climb above or descend below RVSM airspace if the aircraft cannot maintain CFL and ATC cannot establish lateral, longitudinal or conventional vertical separation. - executing the contingency maneuver shown in ENR 1.8 sections 1.2 and 1.3 to offset from the assigned track and FL, if ATC clearance cannot be obtained and the aircraft cannot maintain CFL.	- If the pilot intends to continue in RVSM airspace, assess traffic situation to determine if the aircraft can be accommodated through the provision of lateral, longitudinal, or conventional vertical separation, and if so, apply the appropriate minimum. - If the pilot requests clearance to exit RVSM airspace, accommodate expeditiously, if possible. - If adequate separation cannot be established and it is not possible to comply with the pilot's request for clearance to exit RVSM airspace, advise the pilot of essential traffic information, notify other aircraft in the vicinity and continue to monitor the situation. - Notify adjoining ATC facilities/ sectors of the situation.

Table 2.12.2-2 - Contingency * Scenario # 2	
Loss of redundancy in primary altimetry systems.	
The Pilot should:	ATC can be expected to:
If the remaining altimetry system is functioning normally, couple that system to the automatic altitude control system, notify ATC of the loss of redundancy and maintain vigilance of altitude keeping.	Acknowledge the situation and continue to monitor progress.

Table 2.12.2-3 - Contingency * Scenario # 3	
All primary altimetry systems are considered unreliable or fail.	
The Pilot should:	ATC can be expected to:
Maintain CFL by reference to the standby altimeter (if the aircraft is so equipped). Alert nearby aircraft by - making maximum use of exterior lights; - broadcasting position, FL, and intentions on 121.5 MHZ (as a back-up, the VHF inter-pilot air-to-air frequency, 123.45 MHZ, may be used.)	

Table 2.12.2-3 - Contingency * Scenario # 3	
Consider declaring an emergency. Notify ATC of the failure and intended course of action. Possible courses of action include: - maintaining CFL and route provided that ATC can provide lateral, longitudinal or conventional vertical separation. - requesting ATC clearance to climb above or descend below RVSM airspace if ATC cannot establish adequate separation from other aircraft. - executing the contingency maneuver shown in ENR 1.8 section 1.2 and 1.3 to offset from the assigned track and FL, if ATC clearance cannot be obtained.	Obtain pilot's intentions, and pass essential traffic information. - If the pilot intends to continue in RVSM airspace, assess traffic situation to determine if the aircraft can be accommodated through the provision of lateral, longitudinal, or conventional vertical separation, and if so, apply the appropriate minimum. - If the pilot requests clearance to exit RVSM airspace, accommodate expeditiously, if possible. - If adequate separation cannot be established and it is not possible to comply with the pilot's request for clearance to exit RVSM airspace, advise the pilot of essential traffic information, notify other aircraft in the vicinity and continue to monitor the situation. - Notify adjoining ATC facilities/sectors of the situation.

Table 2.12.2-4 - Contingency * Scenario # 4
The primary altimeters diverge by more than 200 FT (60 M).
The Pilot should:
Attempt to determine the defective system through established trouble-shooting procedures and/or comparing the primary altimeter display to the standby altimeter (as corrected by the correction cards, if required).
If the defective system can be determined, couple the functioning altimeter system to the altitude-keeping device.
If the defective system cannot be determined, follow the guidance in Scenario 3 for failure or unreliable altimeter indications of all primary altimeters.

Table 2.12.2-5 - Contingency * Scenario # 5	
Turbulence (greater than moderate), which the pilot believes will impact the aircraft's capability to maintain flight level.	
The Pilot should:	ATC can be expected to:
Watch for conflicting traffic both visually and by reference to ACAS, if equipped.	
If considered necessary, alert nearby aircraft by: - making maximum use of exterior lights; - broadcasting position, FL, and intentions on 121.5 MHz (as a back-up, the VHF inter-pilot air-to-air frequency, 123.45 MHz, may be used).	
Notify ATC of intended course of action as soon as possible. Possible courses of action include: - maintaining CFL and route provided ATC can provide lateral, longitudinal or conventional vertical separation. - requesting flight level change, if necessary. - executing the contingency maneuver shown in ENR 1.8 section 1.2 and 1.3 to offset from the assigned track and FL, if ATC clearance cannot be obtained and the aircraft cannot maintain CFL.	- Assess traffic situation to determine if the aircraft can be accommodated through the provision of lateral, longitudinal, or conventional vertical separation, and if so, apply the appropriate minimum. - If unable to provide adequate separation, advise the pilot of essential traffic information and request pilot's intentions. - Notify other aircraft in the vicinity and monitor the situation. - Notify adjoining ATC facilities/ sectors of the situation.

2.13 Procedures for LSWD (on the one-way parallel RNAV routes)

2.13.1 LSWD procedures shall be considered when the following conditions exist:

- a. There are at least five (5) weather deviation requests of 10 NM or more within a 30-minute period.
- b. Deviations occur outside radar coverage of the Manila ACC enroute radar.

2.13.2 Flight level assignment for parallel RNAV routes whenever LSWD in effect.

Routes	No-PDC Level Assignment
N892 and M767	FL320, FL360, FL400
L625 and N884 (LAXOR - CAB)	FL310, FL350, FL390
N884 (CAB - LEBIX)	FL290, FL330, FL370, FL410

2.13.3 Other flight levels subject to prior coordination and approval from downstream ACCs.

3. STRATEGIC LATERAL OFFSET PROCEDURES

3.1 Aircraft Navigation Performance and Airspace Safety

3.1.1 Air Traffic Control applies separation minima, including lateral route spacing, based on the assumption that aircraft operate on the center line of a route. In general, unauthorized deviations from this requirement could compromise safety. However, the use of highly accurate navigation systems [such as Global Navigation Satellite System (GNSS)] reduces the magnitude of lateral deviations from the route center line and consequently increases the probability of a collision if a loss of vertical separation between aircraft on the same route occurs.

3.1.2 By using offsets to provide lateral spacing between aircraft, the effect of this reduction in random lateral deviations can be mitigated, thereby reducing the risk of collision.

3.2 Strategic Lateral Offsets in Oceanic Airspace

3.2.1 Offsets are only applied in the oceanic airspace in the Manila FIR.

3.2.2 Offsets are applied only by aircraft with automatic offset tracking capability.

3.2.3 The following requirements apply to the use of the offset:

3.2.3.1 The decision to apply a strategic lateral offset is the responsibility of the flight crew.

3.2.3.2 The offset shall be established at a distance of one (1) or two (2) nautical miles to the right of the center line relative to the direction of flight.

3.2.3.3 The strategic lateral offset procedure has been designed to include offsets to mitigate the effects of wake turbulence of preceding aircraft. If wake turbulence needs to be avoided, one of the three available options (center line, 1 NM or 2 NM right offset) shall be used.

3.2.3.4 In airspace where the use of lateral offsets has been authorized, pilots are not required to inform ATC that an offset is being applied.

3.2.3.5 Aircraft transiting areas of radar coverage in airspace where offset tracking is permitted may initiate or continue an offset.

4. RNP10 AND RNP4 Navigation Requirements

4.1 Introduction

4.1.1 ATC will apply the appropriate lateral separation minima between RNP10 or RNP4 approved aircraft on designated route segments within the Manila FIR as specified in the table below.

ROUTE	SEGMENTS	LATERAL SEPARATION	APPLICABILITY
N892	KABAM TO MIGUG	50 NM	Between RNP10 or RNP4 approved aircraft
L625	ARESI TO MEVIN	50 NM	Between RNP10 or RNP4 approved aircraft
N884	LAXOR TO LUBAN	50 NM	Between RNP10 or between RNP10 and RNP4 approved aircraft
		30 NM	Between RNP4 approved aircraft
M767	JOM TO TEGID	50 NM	Between RNP10 or between RNP10 and RNP4 approved aircraft
		30 NM	Between RNP4 approved aircraft
M772	LAXOR TO ASOBA	50 NM	Between RNP10 or RNP4 approved aircraft

4.1.2 For PBN requirements for other routes, please refer to ENR 3.3 - RNAV Routes.

4.1.3 Pilots must advise ATC of any deterioration or failure of the navigation systems below the navigation requirements for RNP10 or RNP4. ATC shall then provide alternate separation and/or alternative routing.

4.1.4 Aircraft Operators must indicate PBN/A1 in item 18 of the flight plan if meeting RNP10 requirements and PBN/L1 in item 18 of the flight plan if meeting RNP4 requirements.

4.2 Operations by aircraft not meeting RNP10 requirements

4.2.1 An aircraft that is unable to meet the minimum navigation requirement of RNP10 must file flight plan at FL280 or below. Operations above FL280 for these aircraft will be subject to ATC approval, in accordance with the provisions of item 4.2.3 below.

4.2.2 Pilots of such aircraft wishing to operate on ATS routes specified in item 4.1.1, at or above FL290, must indicate their level requirements at Item 18 of the ICAO Flight Plan as RMK/REQ FL (insert level). Approval to operate at the preferred level will be subject to ATC coordination and clearance. Flights that are not approved will be required to operate at FL280 or below or via alternative routes.

4.2.3 ATC units receiving a request from a non-RNP10 approved aircraft to operate on ATS routes specified in item 4.1.1, at or above FL290 will co-ordinate with adjacent ATC units affected by the flight. In deciding whether or not to approve the flight, each ATC unit will take into consideration:

- traffic density;
- communications, including the non-availability of normal communication facilities;
- weather conditions en-route;
- any other factors pertinent at the time.

4.3 Safety assessment criteria

4.3.1 The safety criteria associated with the introduction of the reduced lateral separation minima will be in accordance with the requirements for RNP10 or RNP4 navigation performance, i.e. aircraft navigation performance shall be such that:

- RNP10 - standard deviation of lateral track errors shall be less than 8.7 KM (4.7 NM).
- RNP4 - a lateral navigation accuracy of +/- 7.4 KM (4 NM) for 95% of the flight time.

4.4 Monitoring of aircraft navigation performance

4.4.1 Monitoring of aircraft navigation performance is a joint responsibility between operators, States of Registry or States of Operators (as applicable), regulatory authorities and the ATS providers. The detection and reporting of non-conformance with the navigation requirements against the following parameters will rely primarily on radar monitoring by ATC units.

4.4.1.1 Lateral deviations - a deviation of 15 NM or more from track center line based on radar observations.

4.4.1.2 Longitudinal deviations:

- a. where time separation is being applied by ATC – when the reported separation based on ATC verified pilot estimates vary by three (3) minutes or more from the expected separation at the reporting point; or
- b. where a distance based standard is being applied by ATC based on either ADS, radar observation or RNAV distance reports – when the distance varies by 10 NM or more from the expected distance.

4.4.2 ATC will advise the pilot-in-command when such deviations are observed and implement the required investigation procedures.

4.4.3 The ATC authorities will investigate the causes of such deviations in conjunction with the aircraft operator and the State of Registry, or the State of the Operator, as applicable.

4.5 Separation minima

4.5.1 Lateral Separation Minima

4.5.1.1 A lateral separation minima of 50 NM or 30 NM may only be applied between aircraft equipped in accordance with the provisions of item 4.1.1, operating at FL290 or above, on the ATS route segments mentioned in the same paragraph.

4.5.1.2 When an aircraft not meeting the requirements of RNP10, is approved to operate at or above FL290, on the route segments mentioned in said item, vertical separation shall be applied with aircraft operating on adjacent routes.

4.5.2 Longitudinal Separation

4.5.2.1 A longitudinal separation of 80 NM RNAV or Mach Number Technique (MNT) separation minima may be applied between aircraft equipped in accordance with RNP10 or RNP4 navigation requirements.

4.5.3 Vertical Separation

4.5.3.1 A vertical separation minima of 1000 FT will be applied between aircraft operating between FL290 and FL410 inclusive in accordance with RVSM requirements.

4.6 Operators procedure

4.6.1 The operator shall ensure in-flight procedures, crew manuals and training programs are established in accordance with RNP10 or RNP4 requirements.

4.7 Procedures for the use of RNAV route M772

4.7.1 ATS Route M772 Restrictions

4.7.1.1 RNP10 navigation requirement

- a. Pilots must advise ATC of any deterioration or failure of navigation systems below the navigation requirements for RNP10. ATC shall then provide alternate separation and/or alternate routing.
- b. Pilots of aircraft meeting RNP10 navigation requirements must indicate /R in Item 10 of the ICAO Flight Plan.
- c. An aircraft that is unable to meet the minimum navigation requirements for RNP10 shall not file flight plan to operate on ATS Route M772.

4.7.1.2 Flight planning requirements

4.7.1.2.1 The use of ATS route M772 is subject to the following conditions:

- a. The use of ATS routes M772 for traffic originating from airports in Borneo and landing at Hong Kong (Please see Table 1);
- b. Flight levels 300 and 380 will only be available on M772 under the existing FLAS;
- c. Longitudinal separation of 15 minutes, or 10 minutes based on Mach Number Technique (MNT), shall be applied on M772;
- d. Contingency procedures during LSWD and periods of non-availability of HF communications shall apply within Manila FIR. When LSWD occurs within Manila FIR, Manila ATC may suspend the use of M772, with notification via NOTAM.

Table 1

Departure Aerodrome	Destination Aerodrome
Jakarta - Halim Perdanakusuma (WIIH) - Soekarno Hatta (WIII)	Hong Kong (VHHH) and airports in the People's Republic of China
Kuching (WBGG) Sibu (WBGS) Bintulu (WBGB)	Hong Kong only
Miri (WBGR) Labuan (WBKL) Brunei (WBSB)	Hong Kong only

4.7.1.3 Separation Minima

4.7.1.3.1 Longitudinal Separation

- a. Longitudinal separation minima of 80 NM RNAV or 10 minutes MNT separation minima may be applied between aircraft equipped in accordance with RNP10 navigation requirements.

4.7.1.4 The operator shall ensure that in-flight procedures, crew manuals and training programmes are established in accordance with RNP10 navigation requirements.

4.7.2 In the case of aircraft operating northbound on M772 deviating westward and entering the adjacent airspace of the Ho Chi Minh FIR or the Sanya AOR, the pilot shall obtain an ATC clearance from the Manila ACC to deviate.

4.7.3 In the event that a clearance cannot be obtained in time prior to entering the adjacent airspace, the pilot as a matter of urgency shall contact Ho Chi Minh or Sanya ACC as appropriate, maintain the last cleared flight level and obtain ATC clearance to enter the airspace. Manila Radio shall be informed of the action taken.

4.8 Procedures on RNAV Routes N884 and M767

4.8.1 Introduction

4.8.1.1 This subsection provides additional details relevant to the implementation of 50 NM longitudinal separation between RNP10 approved aircraft and/or RNP4 approved aircraft and 30 NM longitudinal separation minima between RNP4 approved aircraft operating along RNAV routes N884 and M767 within the Manila FIR.

4.8.1.2 Manila Area Control Center (RTF: Manila Control) is the unit providing air traffic control services for all en-route traffic operating within the Manila FIR.

4.8.1.3 RNP10 and RNP4 based longitudinal separation minimum may be applied by ATC between suitably equipped aircraft on RNAV routes N884 and M767 as described in the succeeding paragraphs.

4.8.2 Applicability

4.8.2.1 For aircraft operating on RNAV routes N884 and M767, Performance Based Communication and Surveillance (PBCS) longitudinal separation minima of 50 NM shall be applied between RNP10 approved and/or RNP4 approved aircraft and longitudinal separation minima of 30 NM between RNP4 approved aircraft at FL310, FL320, FL350, FL360, FL390 and FL400.

Note: Please refer to ENR 1.8-6 item 2.2.4 FLAS applicable within the Manila FIR and the adjacent FIRs.

4.8.3 Requirements

4.8.3.1 Aircraft pairs being provided with a reduced longitudinal separation must meet the required PBCS compliance for the application of RNP10 and RNP4 reduced longitudinal separation.

- a. For Philippine registered aircraft, Certification/Approval for RNP operations within the Manila FIR shall be issued by:

Certification and Management Division (CMD)
Flight Safety and Inspectorate Service
Civil Aviation Authority of the Philippines
(02) 7944-2181

- b. Foreign registered aircraft operating within Manila FIR shall have their Certification/Approval for RNP operations from the appropriate agency in their respective States.
- c. The aircraft operator (or designated representative) must indicate the following in the FPL as appropriate.

FLIGHT PLAN			
Capability	Item 10a	Item 10b	Item 18
RNP10	R	-	PBN/A1
RNP4	R	-	PBN/L1
FANS 1/A ADS-C	-	D1	-
FANS 1/A CPDLC	J2-J7	-	-
RCP240	P2	-	COM/RCP240
RSP180	-	-	SUR/RSP180

- 4.8.3.2 To comply with the requirement of Direct Controller-Pilot Communication (DCPC) when applying 50 NM or 30 NM longitudinal separation, pilot must LOG ON to Controller-Pilot Data Link Communication (CPDLC) and utilize VHF radio whenever possible. Such Direct Controller-Pilot Communication (DCPC) shall be maintained at all times when applying these separation minima.
- 4.8.3.3 The communication system should support the application of this minima such that it allows the controller to intervene and resolve a potential conflict:
- Within four (4) minutes, using the normal communication means (VHF radio/CPDLC);
 - Within ten and a half (10 1/2) minutes using an alternate source of communication (HF radio) in the event of normal communication means failure.
- 4.8.3.4 The minima shall be applied only between a traffic pair both logged-on to CPDLC. In other cases, conventional time-based longitudinal separation shall apply, with MNT if deemed appropriate.
- 4.8.3.5 The minima shall only be applied between aircraft on the same track.
- 4.8.3.6 Separation shall be established by maintaining not less than the specified distance between aircraft positions reported through the following means:
- By reference to the same “on track” 1 common point, and whenever possible, ahead of both aircraft (refer to Figures 2 to 4 for calculation of distance between aircraft), or
 - by an automated reporting system.

Figure 2. Aircraft on same identical track

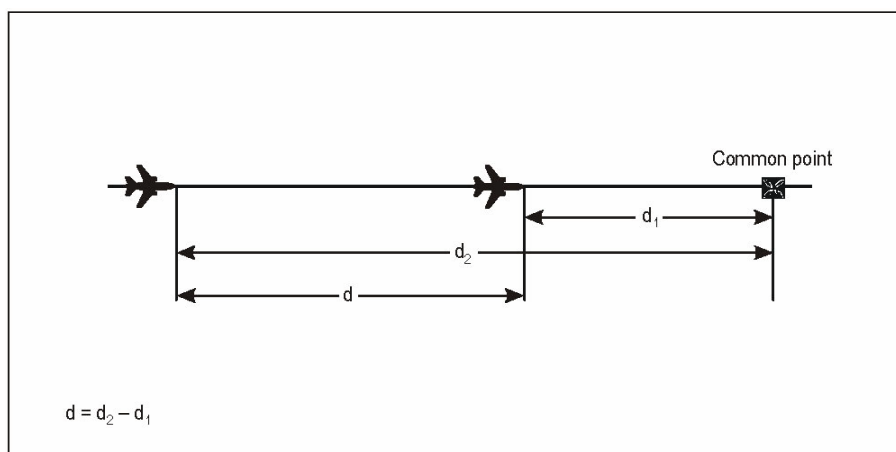


Figure 3. Aircraft on same, but not identical track

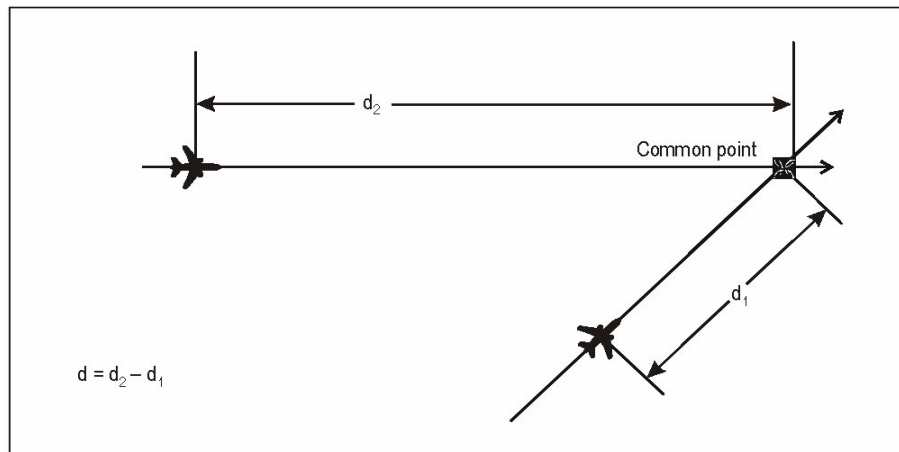
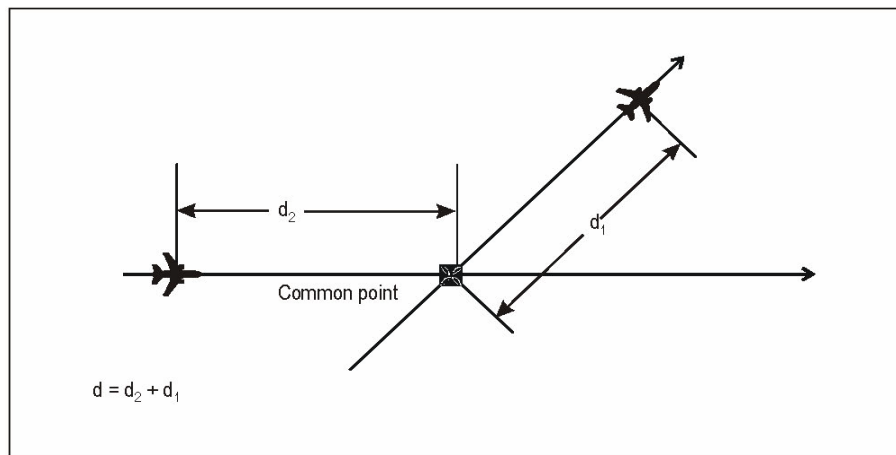


Figure 4. Aircraft on the same track, opposite sides of common point



4.8.3.7 Controllers shall ensure that the minimum distance exists throughout the period of application of the minima. When the aircraft are at or are expected to be at minimum applicable distance, controllers shall apply appropriate speed control techniques, such as assigning appropriate Mach numbers to aircraft.

Note: 1. The term “on track” means that the aircraft is flying either directly inbound to or directly outbound from the station or common point.

2. Same identical tracks are special case of same track where the angular difference is zero degrees.

4.8.3.8 Pilots must report to Manila Control of any deterioration or failure of either of the following:

1. Navigation system (performance below the navigation requirements for RNP10 or RNP4 operations), or
2. ADS-C/CPDLC.

As soon as report has been received, the Manila Control shall provide alternative applicable separation. The change must be coordinated with the unit providing air traffic services in the adjacent FIR.

4.8.4 Application of Longitudinal distance-based separation minima not using ADS-C

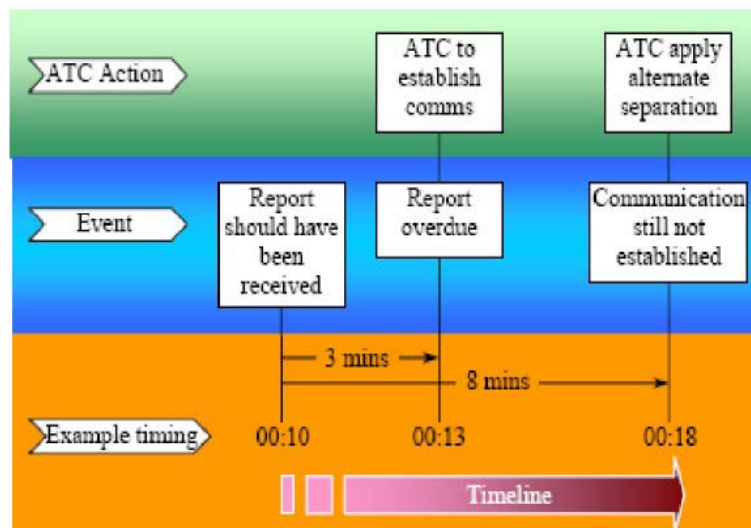
4.8.4.1 The following separation minimum may be used for aircraft climbing, cruising or descending on the same track:

Table 2

Separation Minimum	RNP Type	Communication requirement	Surveillance requirement	Distance verification requirements
50 NM	10	VHF radio or CPDLC DCPC	Position reports or Radar sighting	At least every 24 minutes

4.8.4.2 When an aircraft fails to report its position, the controller shall take action within three (3) minutes to establish communication. If communication is not established within eight (8) minutes from the time the report should have been received, the controller shall apply an alternative form of separation (refer to Figure 5).

Figure 5



4.8.4.3 Controllers are to note that time lag exists between instruction passed and distance reported by pilot, and if left unchecked might result in a misrepresentation of air situation. As such, controllers shall exercise discretion in the sequence of requesting such reports. As a general guide, distance reporting instructions should normally be issued to the front aircraft first.

4.8.4.4 Controller shall use the phraseology "Report distance [to/from] [position]" for aircraft to report its distance to or from a common point ('to' shall be used when position is an ensuing waypoint).

4.8.5 Applicable Longitudinal Separation

4.8.5.1 Within designated airspace, or on designated routes, separation minima in accordance with the provisions may be used.

Separation Minima	RNP	RCP	RSP	Maximum ADS-C periodic reporting interval
50 NM	10	240	180	27 minutes
	4	240	180	32 minutes
30 NM	4	240	180	12 minutes

4.8.5.2 80 NM longitudinal separation between RNAV-approved or RNP10 - approved aircraft without a working CPDLC. A longitudinal separation for ten (10) minutes (or less) may also be applied using the Mach Number Technique (MNT).

4.8.6 Operations by Aircraft Not Meeting RNP 10 Requirements

4.8.6.1 Non-RNP10 - approved aircraft planning to operate along RNAV route N884 (northeast bound) must fly at FL270 or below. Operations above FL270 by these aircraft shall be subject to ATC approval.

4.8.6.2 Whenever a request from a non-RNP10 - approved aircraft to operate on RNAV route N884 (northeast bound) at FL280 or above or on RNAV route M767 (southwest bound) at FL290 or above is received, Manila Control shall coordinate with adjacent ATS units affected by the flight.

4.8.7 Monitoring of Aircraft Navigation Performance

4.8.7.1 The State of registry of the aircraft and of the aircraft operator shall follow the procedures provided in the ICAO Asia-Pacific En-Route Monitoring Agency Handbook for the submission of navigation errors.

4.8.8 Additional Communication Procedures

4.8.8.1 Pilots operating southwest bound along RNAV route M767 may initiate CPDLC communications with Singapore ACC (addressee: WSJC) over waypoint TOKON or twenty (20) minutes before reaching waypoint TEGID but shall remain in contact with Manila Control via VHF at 118.9 MHZ until position TEGID except when instructed by Manila Control to switch to Singapore Control at an earlier time.

4.8.8.2 Pilots operating northeast bound along RNAV route N884 shall establish contact with Manila Control via VHF at 118.9 MHZ at least three (3) minutes before reaching waypoint LAXOR.

4.8.9 Suspension of 50 NM and 30 NM Longitudinal Separation

4.8.9.1 Whenever LSWD procedures are activated along RNAV routes N884 and M767, application of the 50 NM and 30 NM longitudinal separation shall be suspended. Manila Control shall provide other applicable separation.

4.8.9.2 In the event the Philippine Air Traffic Management System encounters system errors which can compromise aircraft position monitoring and/or AGDL communications, 50 NM and 30 NM longitudinal separation shall be suspended. Manila Control shall provide other applicable separation.

a. Suspension shall be coordinated with the concerned ATS units.

b. NOTAM shall be issued.

5. SPECIAL ROUTE AND FLIGHT LEVEL REQUIREMENTS FOR FLIGHTS FROM/TO MANILA FIR AND OVERFLIGHTS DIVERTING TO MANILA FIR

5.1 Manila FIR - Macau Flights

5.1.1 Aircraft departing from airport within Manila FIR bound for Macau International Airport should flight plan to join air route A583 SABNO FL340 or below.

5.2 Mactan - Singapore/Kuala Lumpur and vice versa

5.2.1 Aircraft departing from Mactan-Cebu International Airport should flight plan to join air route:

- a. G578 ZAM A341 VIMAG, or
- b. W17 PR M646 OSANU

5.2.2 Aircraft departing Singapore/Kuala Lumpur Airport bound for Mactan-Cebu International Airport should flight plan to join air route:

- a. VIMAG A341 ZAM G578 MCT, or
- b. OSANU M646 PR W17 MCT

5.3 Flight Level Requirements for Flights From Manila FIR to Ho Chi Minh FIR via PANDI W17 or Vice Versa:

- a. Ho Chi Minh FIR to Manila FIR - FL290, FL270
- b. Manila FIR to Ho Chi Minh FIR - FL280, FL260

Remarks: Minimum of ten (10) minutes MNT longitudinal separation without closing speed will be applied with traffic at PANDI M765 or vice versa.

5.4 To enhance flight safety and to rationalize the flow of traffic transiting Manila FIR from Kota Kinabalu FIR or Singapore FIR to Hong Kong FIR or Taipei FIR and vice versa, flight shall flight plan via:

5.4.1 Kota Kinabalu FIR or Singapore FIR to Hong Kong FIR

- a. VINIK M754 AKOTA A583 SABNO
- b. OSANU M646 MIA CAB A461 NOMAN

5.4.2 Hong Kong FIR to Kota Kinabalu FIR or Singapore FIR

- a. SABNO A583 AKOTA M754 VINIK
- b. NOMAN A461 AVMUP W16 OLIVA MIA M646 OSANU

5.4.3 Kota Kinabalu FIR to Taipei FIR

- a. OSANU M646 POTIB

5.4.4 Taipei FIR to Kota Kinabalu FIR

- a. POTIB M646 OSANU

5.5 During adverse weather, traffic on diversionary route that would transit through Manila FIR from Hong Kong FIR to Ho Chi Minh FIR or beyond and vice versa, flight shall flight plan via:

5.5.1 Hong Kong FIR to Singapore FIR

- a. SABNO A583 AKOTA M754 TENON M767 TEGID

5.5.2 Hong Kong FIR to Ho Chi Minh FIR

- a. SABNO A583 AKOTA M754 NOBEN M765 PANDI

5.5.3 Singapore FIR to Hong Kong FIR

- a. LAXOR N884 LULBU M754 AKOTA A583 SABNO; or
- b. LAXOR N884 LULBU M754 AKOTA L625 AVMUP A461 NOMAN

5.5.4 Ho Chi Minh FIR to Hong Kong FIR

- a. PANDI M765 NOBEN M754 AKOTA A583 SABNO; or
- b. PANDI M765 NOBEN M754 AKOTA L625 AVMUP A461 NOMAN

Note: The following route of flight are no longer permitted:

- a. SABNO A583 MAVRA N892 MIGUG
- b. NOMAN A461 MUMOT N892 MIGUG
- c. ARESI L625 AKOTA A583 SABNO
- d. ARESI L625 AVMUP A461 NOMAN

6. AIR-GROUND COMMUNICATION FOR FLIGHTS FROM HONG KONG FIR INBOUND TO MANILA FIR

RNAV Route/ Waypoint	VHF FREQ (Air-Ground Communication Facility Call Sign)	CPDLC	HF FREQ (Air-Ground Communication Facility Call Sign)
A461/NOMAN	119.3 MHZ (Manila Control)	LOG ON: RPHI	8903 KHZ / 2998 KHZ / 6532 KHZ / 6562 KHZ / 13300 KHZ / 17904 KHZ (Manila Radio)
M501/NOMAN	119.3 MHZ (Manila Control)	LOG ON: RPHI	8903 KHZ / 2998 KHZ / 6532 KHZ / 6562 KHZ / 13300 KHZ / 17904 KHZ (Manila Radio)
A583/SABNO	119.3 MHZ (Manila Control) 5 minutes after SABNO	LOG ON: RPHI	8942 KHZ / 5655 KHZ / 3485 KHZ / 11396 KHZ / 13309 KHZ (Manila Radio)

Note: PRI/SRY HF subject to signal favorability.

7. LONGITUDINAL SEPARATION MINIMA BETWEEN MANILA FIR AND ADJACENT FIRS:

FIR	ATS / RNAV ROUTE	LONGITUDINAL SEPARATION	REMARKS
Taipei	N892	15 minutes	-
		10 minutes	With application of Mach Number Technique (MNT).
	M646	15 minutes	-
		10 minutes	With application of MNT.
		50 NM	One aircraft is landing either Taipei or Manila FIR.

FIR	ATS / RNAV ROUTE	LONGITUDINAL SEPARATION	REMARKS
Fukuoka	B462	15 minutes	-
	N884	10 minutes	Same track or continuously diverging tracks with the application of MNT.
	A582		
	A590	15 minutes	-
		10 minutes	Same track or continuously diverging tracks with the application of MNT.
Oakland	M501	15 minutes	-
	G467	10 minutes	With application of MNT.
	R337		-
Ujung Pandang	G578	15 minutes	-
		10 minutes	With application of MNT
		50 NM	- FL290 and above; - Between RNP10 approved aircraft; and - Separation is maintained or increasing.
	A461	15 minutes	-
		10 minutes	With application of MNT
		50 NM	- FL290 and above; - Between RNP10 approved aircraft; and - Separation is maintained or increasing.
	B472	15 minutes	-
		10 minutes	With application of MNT
		50 NM	- FL290 and above; - Between RNP10 approved aircraft; and - Separation is maintained or increasing.
	R590	15 minutes	-
		10 minutes	With application of MNT
	B473	15 minutes	-
		10 minutes	With application of MNT
	B462	15 minutes	-
		10 minutes	With application of MNT
	A339	15 minutes	-
		10 minutes	With application of MNT
	R342	15 minutes	-
		10 minutes	With application of MNT
Kota Kinabalu	M754	10 minutes	With application of MNT.
	M646	10 minutes / 80 NM	RNAV.
		50 NM	- Between RNP10 approved aircraft and - Separation is maintained or increasing. - Southbound one aircraft landing Kota Kinabalu FIR. - Northbound no restriction on destination.
	A341	10 minutes / 80 NM	RNAV.
		50 NM	- Between RNP10 approved aircraft and - Separation is maintained or increasing. - Westbound one aircraft landing Kota Kinabalu FIR. - Eastbound no restriction on destination.

FIR	ATS / RNAV ROUTE	LONGITUDINAL SEPARATION	REMARKS
Singapore	M767	15 minutes	-
		10 minutes	With application of MNT between RNAV equipped aircraft.
		50 NM	- Between RNP10 approved aircraft and - Separation is maintained or increasing.
		30 NM	- Between RNP4 approved aircraft and - Separation is maintained or increasing.
	N884	15 minutes	-
		10 minutes	With application of MNT between RNAV equipped aircraft.
		50 NM	- Between RNP10 approved aircraft and - Separation is maintained or increasing.
		30 NM	- Between RNP4 approved aircraft and - Separation is maintained or increasing.
	M772	15 minutes	-
		10 minutes	With application of MNT between RNAV equipped aircraft.
Ho Chi Minh	L625	15 minutes	-
		10 minutes	With application of MNT between RNAV equipped aircraft.
	L628	15 minutes	-
		10 minutes	With application of MNT between RNAV equipped aircraft.
	M765	15 minutes	-
		10 minutes	With application of MNT between RNAV equipped aircraft.
	N892	15 minutes	-
		10 minutes	With application of MNT between RNAV equipped aircraft.

FIR	ATS / RNAV ROUTE	LONGITUDINAL SEPARATION	REMARKS
Hong Kong	A461	15 minutes	-
		10 minutes	- With application of MNT or - Irrespective of Mach Number - Between RNAV equipped aircraft - Hong Kong and Manila FIR long-range radars are serviceable and - For flight inbound Hong Kong FIR or for flight inbound Manila FIR provided one aircraft is landing within Manila.
		50 NM	- Between RNP10 approved aircraft and - Separation is maintained or increasing between flight overflying Manila FIR or Hong Kong FIR or for flight inbound Hong Kong FIR or Manila FIR provided one aircraft is landing within Manila or Hong Kong FIR.
		30 NM	- Between RNP4 approved aircraft and - Separation is maintained or increasing between flight overflying Manila FIR or Hong Kong FIR or for flight inbound Hong Kong FIR or Manila FIR provided one aircraft is landing within Manila or Hong Kong FIR.
	A583	15 minutes	-
		10 minutes	- With application of MNT or - Irrespective of Mach Number - Between RNAV equipped aircraft - Hong Kong and Manila FIR long-range radars are serviceable - For flight inbound Hong Kong FIR or for flight inbound Manila FIR provided one aircraft is landing within Manila
		50 NM	- Between RNP 10 approved aircraft and - Separation is maintained or increasing between flight overflying Manila FIR or Hong Kong FIR or for flight inbound Hong Kong FIR or Manila FIR provided one aircraft is landing within Manila or Hong Kong FIR and - CPDLC equipped aircraft pairs.
	M501	15 minutes	-
		10 minutes	- With application of MNT or - Irrespective of Mach Number - Between RNAV equipped aircraft - Hong Kong and Manila FIR long-range radars are serviceable - For flight inbound Hong Kong FIR or for flight inbound Manila FIR provided on aircraft is landing within Manila.
	M772	15 minutes	-
		10 minutes	- With application of MNT or - Irrespective of Mach Number - Between RNAV equipped aircraft - Hong Kong long-range radars are serviceable and - For flight inbound Hong Kong FIR.

8. AIR TRAFFIC MANAGEMENT CONTINGENCY PLAN

8.1 Introduction

8.1.1 In compliance with the requirements of the ICAO Standards and Recommended Practices contained in Annex 11 and the Regional Supplementary Procedures (Doc 7030), the Air Traffic Management (ATM) Contingency Plan for Manila FIR has been developed. The contingency plan provides for the contingency arrangements to allow the continuance of international flights to transit the Manila FIR during partial or total disruption in the normal Air Traffic Control and Support Services provided by Philippine ATS.

8.1.2 This contingency plan does not address arrangements for aircraft arriving and departing at Philippine airports.

This ATM Contingency Plan provides:

1. ATS procedures relevant to the contingency plan; and
2. the contingency routes structure using existing published airways to allow transit through the Manila FIR.

8.1.3 As and where dictated by circumstances, aircraft planning to operate through Manila FIR that have not yet departed may be temporarily suspended until a full assessment of the prevailing conditions has been determined and sufficient air traffic services restored.

8.1.4 Long-haul international aircraft and special operations (e.g. Search and Rescue (SAR), State aircraft, humanitarian flights, etc.), shall be afforded priority for levels at FL290 and above.

8.1.5 International operators may elect to avoid the Manila FIR to the east routing via Ujung Pandang, Oakland and Fukuoka FIRs; to the west routing via Ujung Pandang, Kota Kinabalu, Singapore, Ho Chi Minh to Hong Kong and Taipei FIRs. The contingency routes to be used in this scenario will be provided by the ATS authorities concerned.

8.2 Reduced ATS and Provision of Flight Information Services (FIS)

8.2.1 During the contingency critical period, ATS including ATC services may not be available, particularly with regard to availability of communications and radar services. In cases where service is not available, a NOTAM will be issued providing the relevant information. The contingency plan provides for limited flight information and alerting services to be provided by adjacent ACCs.

8.2.2 FIS and flight monitoring will be provided by the designated ATS authorities for the adjacent FIRs on the contingency routes that enter their respective FIRs.

8.2.3 During the early stages of a contingency event, ATC may be overloaded and tactical action may be taken to reroute aircraft on alternative routes not included in this plan.

In the event that ATS cannot be provided in the Manila FIR, a NOTAM shall be issued indicating the following:

- a. time and date of the commencement of contingency measures;
- b. airspace available for overflying traffic;
- c. details of the facilities and services available and limitations on ATS provision (e.g. ACC, APPROACH, TOWER and FIS), including an expected date of restoration of services if available;
- d. information on the provisions made for alternative services;
- e. any changes to the ATS contingency routes contained in this plan;

- f. any special procedures to be followed by neighboring ATS units not covered by this plan;
- g. any special procedures to be followed by pilots; and
- h. any other details that aircraft operators may find useful with respect to the disruption and actions taken.

8.3 Aircraft Separation and Spacing

- 8.3.1 Aircraft separation criteria will be applied in accordance with the ICAO Procedures for Air Navigation Services - Air Traffic Management (PANS-ATM, Doc 4444) and the Regional Supplementary Procedures (Doc 7030).
- 8.3.2 The longitudinal separation/spacing will be fifteen (15) minutes. However, this may be reduced to ten (10) minutes in conjunction with the application of the Mach number technique (MNT).

8.4 Priority for Flight Levels

- 8.4.1 Where possible, aircraft on long-haul international flights shall be afforded priority for cruising levels assigned in accordance with the FLAS as specified in Table 3.

8.5 Aircraft Position Reporting

- 8.5.1 Beyond VHF coverage, Automatic Dependent Surveillance - Contract (ADS-C) shall replace any requirement for voice position reporting to ATC for suitably equipped aircraft and in this case Controller-Pilot Data Link Communications (CPDLC) or HF will be the secondary means of communication. When CPDLC has been authorized for use by the relevant ATC authority, this will become the primary means of communication while HF will act as the secondary means of communication.

8.6 Exclusions

- 8.6.1 VFR flights shall not operate in the Manila FIR during contingency operations, except for State aircraft, Medevac flights and any other aircraft as authorized by CAAP.

8.7 Pilots and Operator Procedures

- 8.7.1 Filing of flight plans

Flight planning requirements detailed in Philippine AIP continue to apply during contingency operations, except where modified by the contingency ATS routes and FLAS specified by ATC and/or in NOTAM.
- 8.7.1.1 Airspace users are expected to familiarize themselves with the Contingency Plan of the Manila FIR and the activation times. For aircraft intending to operate in areas during periods when the Contingency Plan is activated, the operators shall plan the flight to conform to the requirements of the Contingency Plan.
- 8.7.1.2 The flight planning requirements during contingency periods will be in accordance to ICAO Annex 2 Chapter 3 and Doc 4444 Chapter 4 and Appendix 2. Additional information will, however, be required to indicate that the aircraft will operate in airspace where the Contingency Plan is active.
- 8.7.2 Overflight approval
- 8.7.2.1 Airspace users must obtain overflight approval from CAAP prior to operating aircraft through the Manila FIR.

8.7.3 Pilot operating procedures

8.7.3.1 Pilot of an aircraft operating in the Manila FIR during contingency operations shall comply with the following procedures:

- a. all aircraft proceeding along the ATS routes established in this Contingency Plan will comply with the Instrument Flight Rules (IFR) and will be assigned a flight level in accordance with the FLAS applicable to the route(s) being flown as specified in Table 3;
- b. flights are to file flight plan using the Contingency Routes specified in Table 3, according to their airport of origin and destination;
- c. pilots are to keep a continuous watch on the specified contingency frequency as specified in Table 3 and transmit in English the position information and estimates in line with normal ATC position reporting procedures;
- d. pilots are to maintain during their entire flight time within Manila FIR, the flight level last assigned by the last ACC responsible prior to the aircraft entering the Manila FIR, and under no circumstances change this level and Mach number, except in cases of emergency and for flight safety reasons. In addition, the last SSR transponder assigned shall be maintained or, if no transponder has been assigned, transmit on SSR code 2000;
- e. aircraft are to reach the flight level last assigned by the responsible ACC at least 10 minutes before entering the Manila FIR;
- f. pilots are to include in their last position report prior to entering the Manila FIR, the estimates over the entry point, the OCA agreed Transfer of Control Point (TCP) and the ETA at the exit point;
- g. pilots should establish communication with the next ATC facility at least 10 minutes prior to the OCA agreed TCP;
- h. whenever emergencies and/or flight safety reasons make it impossible to maintain the flight level assigned for transit of Manila FIR, pilots are to climb or descend well to the right of the center line of the contingency route, and if deviating outside the Manila FIR, to inform immediately the ACC responsible for that airspace. Pilots are to make blind broadcast on 121.5 MHZ of the relevant emergency level change message (comprising the aircraft call sign, the aircraft position, the flight levels being vacated and crossed, etc); and
- i. not all operational circumstances can be addressed by this Contingency Plan and pilots are to maintain a high level of alertness when operating in the contingency airspace and take appropriate action to ensure safety of flight.

8.7.4 Interception of Civil Aircraft

8.7.4.1 Aircraft operators must be familiar with international intercept procedures contained in ICAO Annex 2 - Rules of the Air, Chapter 3, paragraph 3.8 and Appendix 2, sections 2 and 3.

8.7.4.2 Pilots are to comply with instructions given by the pilot of the intercepting aircraft. In such circumstances, the pilot of the aircraft being intercepted shall broadcast information on the situation.

8.7.4.3 If circumstances leading to the closure of the Manila FIR where no contingency routes are available, aircraft will be required to keep clear of Manila FIR. As much warning as possible will be provided by the appropriate ATS authorities in the event of the complete closure of airspace.

8.7.4.4 Pilots shall continuously guard the VHF emergency frequency 121.5 MHZ and shall operate their transponder at all times during flight, regardless of whether the aircraft is within or outside airspace where secondary surveillance radar (SSR) is used for ATS purposes. Transponders should be set on the last discrete code assigned by ATC or select Code 2000 if no code was assigned.

8.8 Communication Procedures

8.8.1 Degradation of Communication - Pilot Radio Procedures

8.8.1.1 When operating within the contingency airspace of the Manila FIR, pilots should use normal radio communication procedures where ATS services are available. These will be in accordance with the communication procedures in this plan or as otherwise notified by NOTAM.

8.8.1.2 If communications are lost unexpectedly on the normal ATS frequencies, pilots shall try the next applicable frequency, e.g. if en-route contact is lost then try the next appropriate frequency, that is, the next normal handover frequency. Pilots should also consider attempting to contact ATC on the last frequency where two-way communication had been established. In the absence of communication with ATC, the pilot shall continue to make routine position reports on the assigned frequency, and also broadcast positions on the specified contingency frequency.

8.8.2 Communication Frequencies

8.8.2.1 A list of frequencies to be used for the contingency routes and the ATS units providing FIS and air-ground communication monitoring for the Manila FIR is detailed in Table 3.

8.9 Level 2 - International Contingency Routes of Manila FIR

Table 3: Contingency Route Structure, Flight Level Allocation Scheme (FLAS) and Transfer of Communication Procedures

Contingency Route	ATS Route	Direction	FLAS	ATS Unit	Transfer of Communication	Remarks
CRM 1	N892 KABAM - MIGUG	Southwest	FL320 FL360 FL400	Taipei ACC	At MUMOT, log on to VVHM for ADS-C CPDLC equipped aircraft. At MIGUG, contact Ho Chi Minh ACC 120.7 MHZ.	International operators may choose to avoid the Manila FIR by using alternate ATS routes in other FIRs.
CRM 2	L625 - M646 ARESI - POTIB	Northeast	FL310 FL350 FL390	Ho Chi Minh ACC	Applicable between 1400 - 2200. Contact Taipei ACC 129.1 MHZ or 127.9 MHZ five (5) minutes before POTIB. Applicable between 2200 - 1400. At AVMUP, contact Laoag APP 122.3 MHZ.	International operators may choose to avoid the Manila FIR by using alternate ATS routes in other FIRs. Laoag APP Facility hours of operations: 2200 - 1400.
				Laoag APP	Laoag APP to advise aircraft, contact Taipei ACC 129.1 MHZ or 127.9 MHZ five (5) minutes before POTIB.	
	M646 POTIB - ABVAR	Southwest	Nil	Taipei ACC	Nil	International operators are restricted from entering the Manila FIR from the Taipei FIR via POTIB M646.

Table 3: Contingency Route Structure, Flight Level Allocation Scheme (FLAS) and Transfer of Communication Procedures

Contingency Route	ATS Route	Direction	FLAS	ATS Unit	Transfer of Communication	Remarks
CRM 3	N884 LAXOR - LULBU	Northeast	FL310 FL350 FL390	Singapore ACC	Applicable between 1300 - 2100. At LUBAN, contact Clark APP 119.2 MHZ. Applicable between 2100 - 1300. At LULBU, contact Puerto Princesa APP 122.0 MHZ.	International operators may choose to avoid the Manila FIR by using alternate ATS routes in other FIRs. Puerto Princesa APP Facility hours of operations: 2100 - 1300.
	N884 LULBU - LUBAN			Puerto Princesa APP	Puerto Princesa APP to advise aircraft at LUBAN, contact Clark APP 119.2 MHZ.	Clark APP Facility hours of operations: H24.
	N884 LUBAN - DADNU			Clark APP	Applicable between 1400 - 2200. Clark APP to advise aircraft at DADNU, contact Tokyo Radio on HF*. Applicable between 2200 - 1400. Clark APP to advise aircraft at DADNU, contact Laoag APP 122.3 MHZ.	Laoag APP Facility hours of operations: 2200 - 1400. *HF will be notified by Fukuoka ACC.
	N884 DADNU - LEBIX			Laoag APP	Laoag APP to advise aircraft at 20 NM South of LEBIX (2040N on N884), contact Fukuoka ACC 123.9 MHZ.	

Table 3: Contingency Route Structure, Flight Level Allocation Scheme (FLAS) and Transfer of Communication Procedures

Contingency Route	ATS Route	Direction	FLAS	ATS Unit	Transfer of Communication	Remarks
CRM 4	B462 MEVIN - LAO	Southwest	FL320 FL360 FL400	Fukuoka ACC	Applicable between 1400 - 2200. At 40 NM North of CAB (1609N on B462), contact Clark APP 119.2 MHZ. Applicable between 2200 - 1400. At 130 NM northeast of LAO (2000N on B462), contact Laoag APP 122.3 MHZ.	International operators may choose to avoid the Manila FIR by using alternate ATS routes in other FIRs. Laoag APP Facility hours of operations: 2200 - 1400. Clark APP Facility hours of operations: H24.
	B462 LAO - CAB			Laoag APP	Applicable between 2200 - 1400. Laoag APP to advise aircraft at 40 NM North of CAB (1609N on B462), contact Clark APP 119.2 MHZ.	Puerto Princesa APP Facility hours of operations: 2100 - 1300.
	CAB - MIA - VERDE - TAPAP			Clark APP	Applicable between 1300 - 2100. Clark APP to advise aircraft at TAPAP, log on WSJC for ADS-C CPDLC equipped aircraft or contact Singapore HF: 6556 / 8942 KHZ. Applicable between 2100 - 1300. Clark APP to advise aircraft at TOKON, contact Puerto Princesa APP 122.0 MHZ.	
	M767 TAPAP - TEGID			Puerto Princesa APP	Applicable between 2100 - 1300. Puerto Princesa APP to advise aircraft at TOSOV, log on to WSJC for ADS-C CPDLC equipped aircraft or contact Singapore HF: 6556 / 8942 KHZ.	

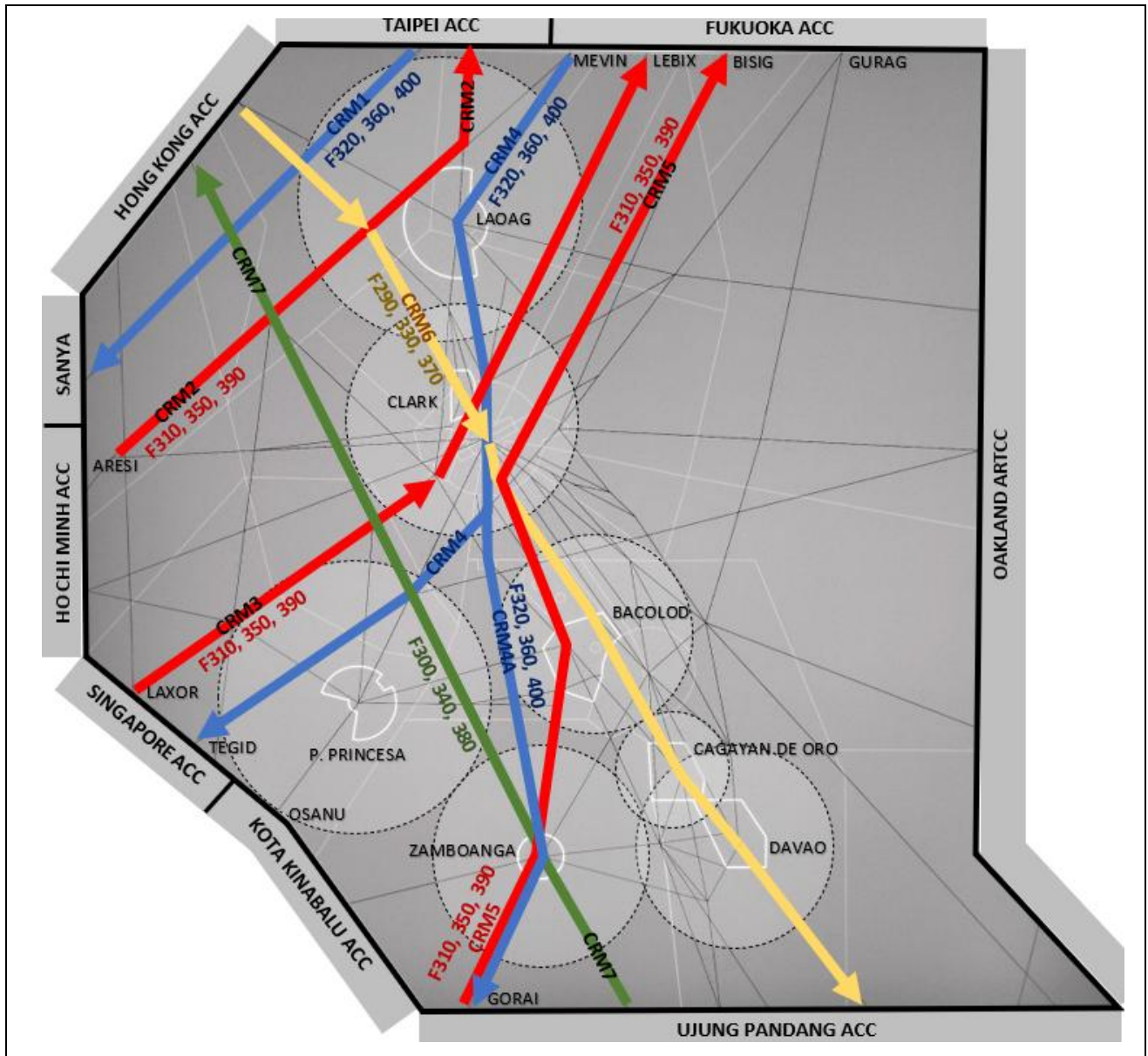
Table 3: Contingency Route Structure, Flight Level Allocation Scheme (FLAS) and Transfer of Communication Procedures

Contingency Route	ATS Route	Direction	FLAS	ATS Unit	Transfer of Communication	Remarks
CRM 4A	A461 TAPAP - ZAM	Southwest	FL320 FL360 FL400	Clark APP	Clark APP to advise aircraft at PUSIT, contact Bacolod APP 121.0 MHZ.	International operators may choose to avoid the Manila FIR by using alternate ATS routes in other FIRs. Zamboanga APP Facility hours of operations: 2100 - 1300.
				Bacolod APP	Applicable between 1300 - 2100. Bacolod APP to advise aircraft at GORAI, contact Ujung Pandang ACC 132.5 MHZ. Applicable between 2100 - 1300. Bacolod APP to advise aircraft at TINDO, contact Zamboanga APP 122.7 MHZ.	
	G578 ZAM - GORAI			Zamboanga APP	Zamboanga APP to advise aircraft at TABUL, log on to WAAF for ADS-C CPDLC equipped aircraft or at GORAI, contact Ujung Pandang ACC 132.5 MHZ.	
CRM 5	G578 GORAI - ZAM	Northeast	FL310 FL350 FL390	Ujung Pandang ACC	Applicable between 1300 - 2100. Ujung Pandang ACC to advise aircraft at GAYUD, contact Bacolod APP 121.0 MHZ. Applicable between 2100 - 1300. Ujung Pandang ACC to advise aircraft at TABUL, contact Zamboanga APP 122.7 MHZ.	International operators may choose to avoid the Manila FIR by using alternate ATS routes in other FIRs. Zamboanga APP Facility hours of operations: 2100 - 1300. *HF will be notified by Fukuoka ACC.
				Zamboanga APP	Zamboanga APP to advise aircraft at GAYUD, contact Bacolod APP 121.0 MHZ.	
				Bacolod APP	Bacolod APP to advise aircraft at NUMRU, contact Clark APP 119.2 MHZ.	
	ZAM - W3 - IOO - B472 - NUMRU - A582 - BISIG			Clark APP	Clark APP to advise aircraft at PIBEM, contact Tokyo Radio on HF*.	

Table 3: Contingency Route Structure, Flight Level Allocation Scheme (FLAS) and Transfer of Communication Procedures

Contingency Route	ATS Route	Direction	FLAS	ATS Unit	Transfer of Communication	Remarks
CRM 6	NOMAN A461 - AVMUP - W16 - TATIL	Southeast	FL290 FL330 FL370	Hong Kong Air Traffic Control Center (HKATCC)	Applicable between 1400 - 2200. At NOMAN, HKATCC to advise aircraft to contact Clark APP 119.2 MHZ at 100 NM before CIA DVOR/DME. Applicable between 2200 - 1400. At NOMAN, HKATCC to advise aircraft to contact Laoag APP 122.3 MHZ at AVMUP.	International operators may choose to avoid the Manila FIR by using alternate ATS routes in other FIRs. Laoag APP Facility hours of operations: 2200 - 1400. Clark APP Facility hours of operations: H24.
				Laoag APP	Laoag APP to advise aircraft at 100 NM to CIA DVOR/DME, contact Clark APP 119.2 MHZ.	Bacolod APP Facility hours of operations: H24.
				Clark APP	Clark APP to advise aircraft at MINOR, contact Bacolod APP 121.0 MHZ.	Laguindingan APP Facility hours of operations: H24. Davao APP Facility hours of operations: H24.
	MINOR - B473 - ELNIT			Bacolod APP	Bacolod APP to advise aircraft at LOWAY, contact Laguindingan APP 122.3 MHZ.	
				Laguindingan APP	Laguindingan APP to advise aircraft at ANOXI, contact Davao APP 122.4 MHZ.	
				Davao APP	Davao APP to advise aircraft at ELNIT, log on WAAF for ADS-C CPDLC equipped aircraft, or contact Ujung Pandang ACC 128.1 MHZ.	
CRM 7	NITIP - A461 - ZAM - A583 - SABNO	Northwest	FL300 FL340 FL380	Ujung Pandang ACC	Applicable between 1300 - 2100. To contact Hong Kong Radar 128.75 MHZ five (5) minutes before SABNO. Applicable between 2100 - 1300. At NITIP, contact Zamboanga APP 122.7 MHZ.	International operators may choose to avoid the Manila FIR by using alternate ATS routes in other FIRs. Zamboanga APP/TWR Facility hours of operations: 2100 - 1300. Puerto Princesa APP Facility hours of operations: 2100 - 1300.

Figure 6. Manila ACC Contingency Route Structure Chart



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